

Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 13, 1999

Instructions: Answer all four questions. Open book and open notes.

1. (a) Two cards are randomly drawn from a deck of 52 playing cards. Find the probability that both cards will be strictly greater than 3 and strictly less than 8.

(b) It is felt that the probabilities are .3, .4, .3 that the teams of three universities T , U , V will win a championship. If U is placed on probation what is the probability that V will win?

2. Find the cumulative distribution of the random variable which has probability distribution

$$f(x) = \frac{x}{15} \quad \text{for } x = 1, 2, 3, 4, 5.$$

3. Given X is a binomial random variable with parameters (n, p) . I.e.

$$f_X(x; p) = C_x^n p^x (1-p)^{n-x}.$$

(a) Show whether $\frac{X+1}{n+2}$ is an unbiased estimator of p .

(b) Is it asymptotically unbiased, i.e. as $n \rightarrow \infty$, does the bias go to zero?

4. A single observation is taken from an exponential distribution with parameter θ ;
i.e.

$$f_X(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0.$$

Test $H_0 : \theta = 2$ vs $H_1 : \theta = 5$. Consider the test accept H_0 if $X < 3$. Find the probabilities of type I and type II errors.

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 13, 1999

Instructions: Answer three out of four questions. Open book and open notes.

1. The Allan Preparation Course is designed to help students achieve better scores on the SAT exam. The accompanying table lists results for randomly selected students. Use the indicated test with a 0.05 level of significance to test the hypothesis that the course has no effect on SAT scores.

- (a) Sign test
- (b) Wilcoxon signed-rank test
- (c) t test for a claim about two dependent samples
- (d) How do the preceding results support the statement that nonparametric tests lack sensitivity of parametric tests (so stronger evidence is required before a null hypothesis is rejected)?

Subject	A	B	C	D	E	F	G	H	I	J
SAT Score before course	700	840	830	860	840	690	830	1180	930	1070
SAT score after course	800	840	820	980	980	800	800	1270	1080	1220

2. Here are the summary data from an experiment involving a treatment with four levels:

	Treatment			
	A1	A2	A3	A4
n	8	8	8	8
$\sum y_i$	28	22	62	50

Source	SS	df
Treatment	131.375	3
Error	41.0	28

- (a) Complete the ANOVA table.
- (b) Is there a significant difference in the treatment means?
- (c) Analyze the means using Tukey's multiple comparison procedure with $\alpha = .01$.

3. The following data were obtained in a study of the relation between diastolic blood pressure (Y) and age (X) for boys 5 to 13 years old.

$i:$	1	2	3	4	5	6	7	8
$X_i:$	5	8	11	7	13	12	12	6
$Y_i:$	63	67	74	64	75	69	90	60

$$\bar{X} = 9.25, \bar{Y} = 70.25, \sum X_i^2 = 752, \sum Y_i^2 = 40,116, \sum X_i Y_i = 5356$$

(a) Assuming normal error regression model is appropriate, obtain the estimated regression function and plot the residuals e_i against X_i . What does your residual plot show?

(b) Omit case 7 from the data and obtain the estimated regression function based on the remaining seven cases. Compare this estimated regression function to that obtained in part (a). What can you conclude about the effect of case 7?

(c) Using your fitted regression function in part (b), obtain a 99 percent prediction interval for a new Y observation at $X = 12$. Does observation Y_7 fall outside this prediction interval? What is the significance of this?

4. The data in the accompanying table were obtained through a survey of randomly selected subjects.

(a) If one of the survey subjects is randomly selected, find the probability of getting someone ticketed for speeding.

(b) If one of the survey subjects is randomly selected, find the probability of getting a male or someone ticketed for speeding.

(c) Find the probability of getting someone ticketed for speeding, given that the selected person is a man.

(d) Find the probability of getting someone ticketed for speeding, given that the selected person is a woman.

(e) Use a 0.05 level of significance to test the claim that the percentage of women ticketed for speeding is less than the percentage of men. Can we conclude that men generally speed more than women?

	Ticketed for Speeding Within the Last Year	
	Yes	No
Men	26	224
Women	27	473

[Based on data from R.H. Bruskin Associates]